

# **YTRS0289**

## **Quick Start Guide**

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Version : 1.2

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# 1 Introduction

## YTRS



### **Features:**

#### **RF Capabilities**

- Based on liquid crystal
- 20x20 unit cells
- 29.2GHz to 29.6GHz frequency band
- 100MHz bandwidth
- Optional module scalable

#### **Control**

- Xilinx FPGA

#### **Software**

- MATLAB, GUI

#### **Peripherals**

- DCI Interface
- 2.54mm pin header

#### **Power**

- 12VDC, 300mA

#### **Outline**

- 133mm x 154mm x 54.2mm

#### **Applications:**

- Applicate 6G and academic research

## 2 Verifying the Kit Contents

### 2.1 Kit Contents

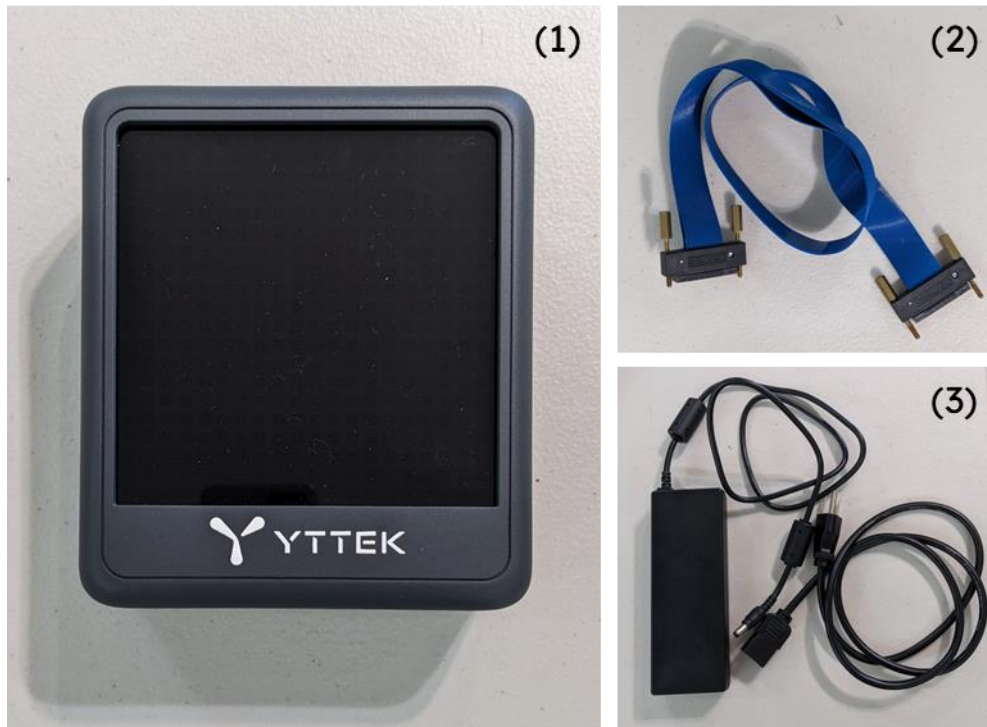


Figure 2-1: YTRS0289 and Accessories

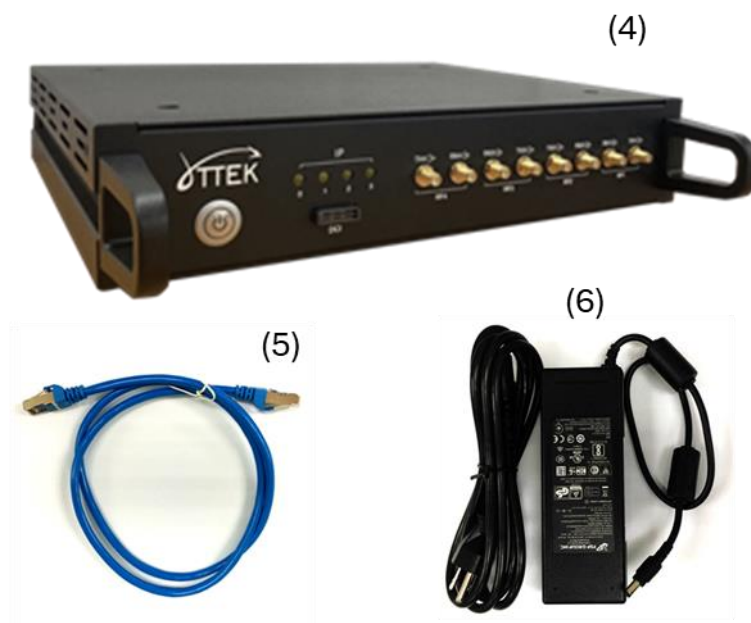


Figure 2-2: Y-FORCE and Accessories

The kit contents of YTRS0289 and Y-FORCE are as follows

1. YTRS0289
  2. 60-pin DCI cable
  3. DC Power adapter 12V / 7A with 110V power cord
  4. Y-FORCE Series
  5. Ethernet Cable
  6. DC Power adapter 12V / 7A with 110V power cord
- ※ Need a Y-FORCE platform to control the beam steering of the YTRS0289 with the DCI cable.

#### ● Preparing the Environment

Ensure that the environment users are using the YTRS0289 in meets the following specifications.

|                                 |                                              |
|---------------------------------|----------------------------------------------|
| Ambient temperature range       | 0°C to 50°C                                  |
| Optimized Operating temperature | 23 °C ± 5 °C                                 |
| Operating humidity              | 20% to 80% relative humidity, noncondensing  |
| Indoor use only.                | The platform is not waterproof and dustproof |



Do not operate the YTRS0289 platform in an environment not specified as above. Product misuse can result in a hazard.



If the product is damaged, return it to YTTEK for repair. It is not recommended to repair yourself

## 2.2 YTRS0289 Setup

### DCI Control



### UART Control

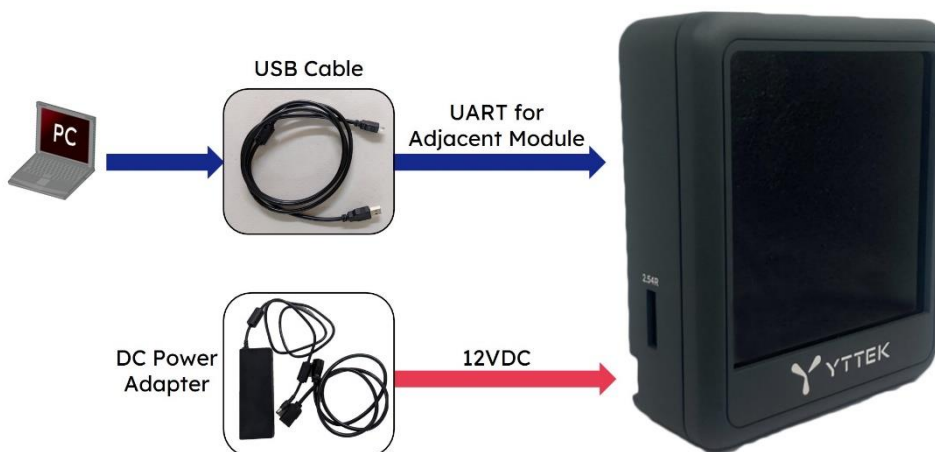


Figure 2-3: YTRS0289 Setup

## 2.3 YTRS0289 Function & Interface

1. Use the DCI cable to connect the YTPC0289 to the Y-FORCE series. Please do not unplug the cable during operation.
2. Connect the 12V DC power jack to the YTRS0289.
3. Connect the 110V power cord.

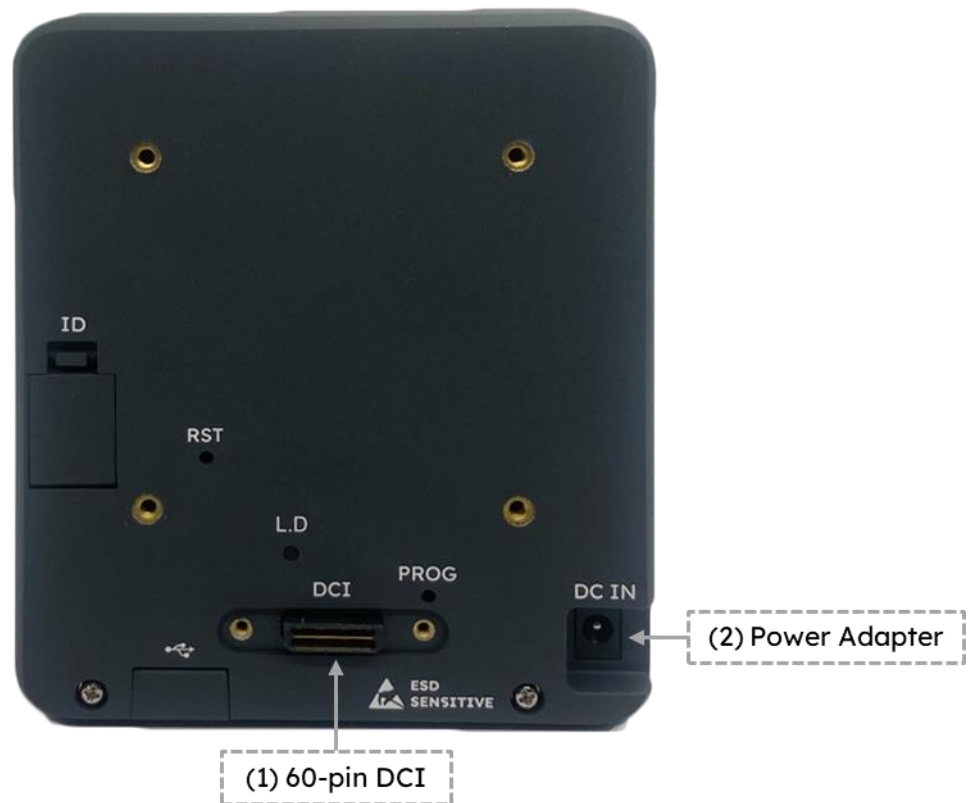


Figure 2-4: YTRS0289 DCI & Power Interface

### ● Uninstall YTRS0289

1. Unplug the 110V power cord.
2. Disconnect the 12V DC power jack.
3. Unplug the DCI cable.



(a) Front side



(b) Right side



(c) Rear side



(d) Left side

Figure 2-5: YTRS0289 function & interface

Table 2-1: YTRS0289 functions

| Item     | Description                                                                                                                                                                     |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LC-Panel | A liquid crystal-based panel provides reflected beams which have arbitrary direction by applying different voltage distribution.                                                |
| DCI      | A digital control signal interface that connects to the YTTEK Y-FORCE series for beam-forming realization.                                                                      |
| DC IN    | A power supply interface that connects to the AC to DC adapter which has 12V and 7A(max) power handling.                                                                        |
| 2.54R/L  | Two 16-pin headers that cascade the power and control signal to the adjacent YTRS0289. Provide the function that extends the aperture size by connecting plurality of YTRS0289. |
| L.D      | A led shows the indicator when FPGA is ready.                                                                                                                                   |
| RST      | A tactile switch that can reset FPGA when FPGA is in an abnormal state.                                                                                                         |
| ID       | A DIP switch that has 8 positions assigns the numbering of YTRS0289. Each YTRS0289 has a different numbering when plurality of YTRS0289 is serial connected.                    |
| USB      | A mini-USB interface that connects to PC for beam-forming realization.                                                                                                          |
| PROG     | For internal engineering debug use only.                                                                                                                                        |

## 3 Platform Software Quick Start

### 3.1 Software Environment Setup

1. Download and install the **MATLAB Runtime version 9.12 (R2022a)** from below address: <https://www.mathworks.com/products/compiler/matlab-runtime.html>
2. Download the folder: “**.\release**” which includes YTRS0289 API and GUI program from below address:  
[https://drive.google.com/drive/folders/1zvIOM8ARaRIbGs9jhKTqjSmsJqWmpI3u?usp=drive\\_link](https://drive.google.com/drive/folders/1zvIOM8ARaRIbGs9jhKTqjSmsJqWmpI3u?usp=drive_link)
3. **(DCI Control)** Configure the host Ethernet interface with a static IP address, and set the IP address “**192.168.3.200**” on the host computer. If users use Ethernet adapter, please use “USB 3.0 to gigabit Ethernet adapter”. Following the steps:
  - I. Choose the network adapter which is connected to the platform. Right click on the network adapter and choose “Properties”.

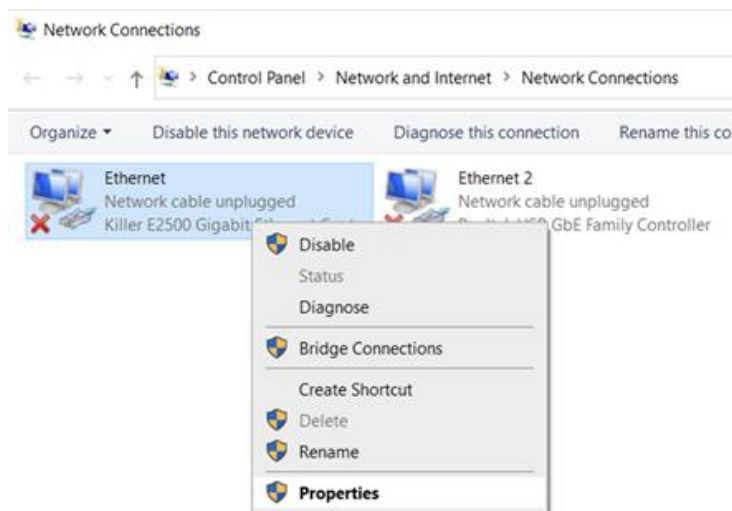


Figure 3-1: Network Connections

- II. Choosing the Internet Protocol Version 4 (TCP/IP). It is better to cancel all of the ticks in the menu except Internet Protocol Version 4 (TCP/IP).

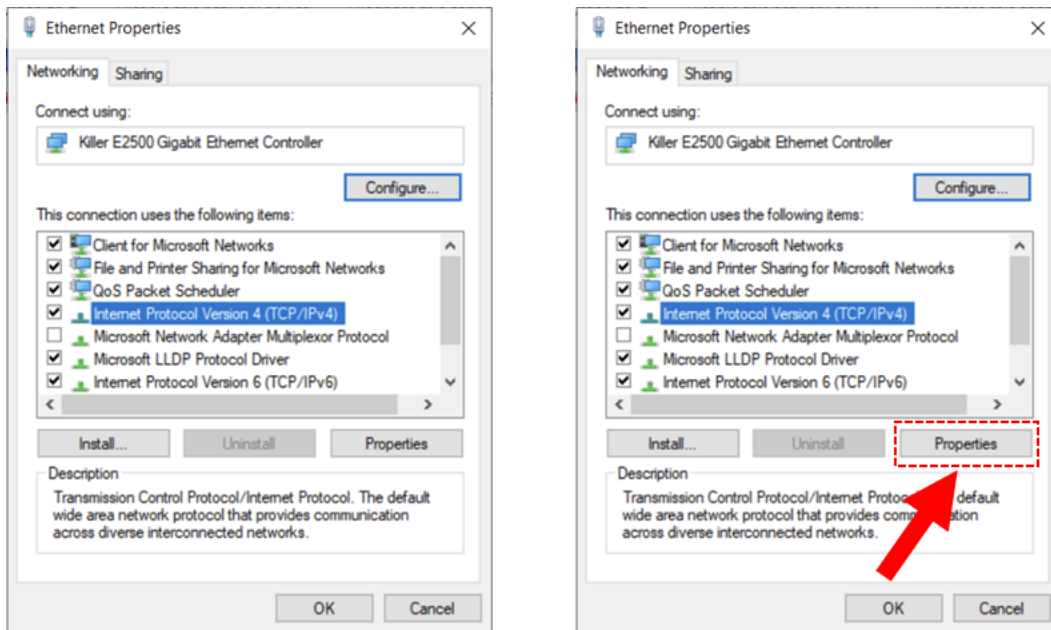


Figure 3-2: Ethernet Properties

- III. Tick the label “using the following IP Address”. Fill the IP Address with 192.168.3.200, and sub-mask with 255.255.255.0., click OK and finish IP setting process.

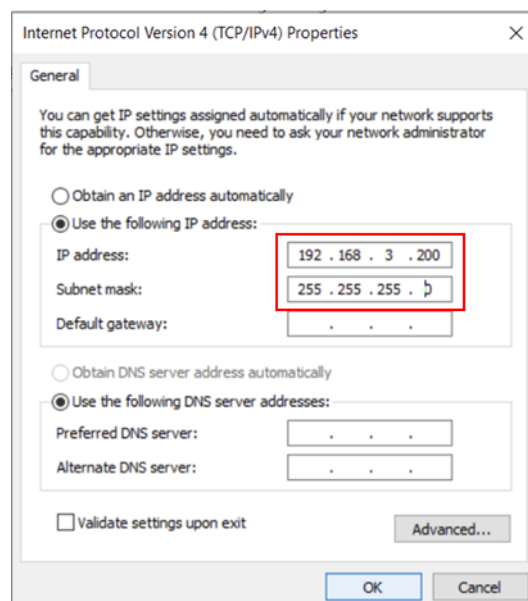


Figure 3-3: IP4 Properties

#### IV. Turn off the firewall to prevent the firewall from blocking network packets.

##### Customize settings for each type of network

You can modify the firewall settings for each type of network that you use.

###### Private network settings



☐ Turn on Windows Defender Firewall

☐ Block all incoming connections, including those in the list of allowed apps

☒ Notify me when Windows Defender Firewall blocks a new app



☒ Turn off Windows Defender Firewall (not recommended)

###### Public network settings



☐ Turn on Windows Defender Firewall

☐ Block all incoming connections, including those in the list of allowed apps

☒ Notify me when Windows Defender Firewall blocks a new app



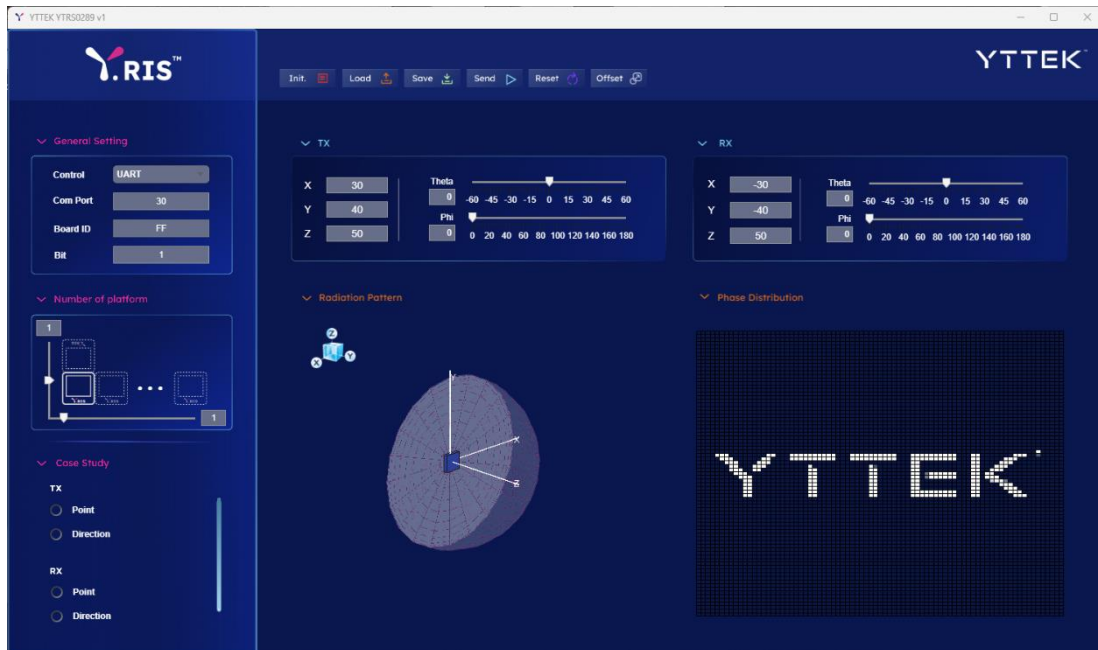
☒ Turn off Windows Defender Firewall (not recommended)

Figure 3-4: Windows Firewall

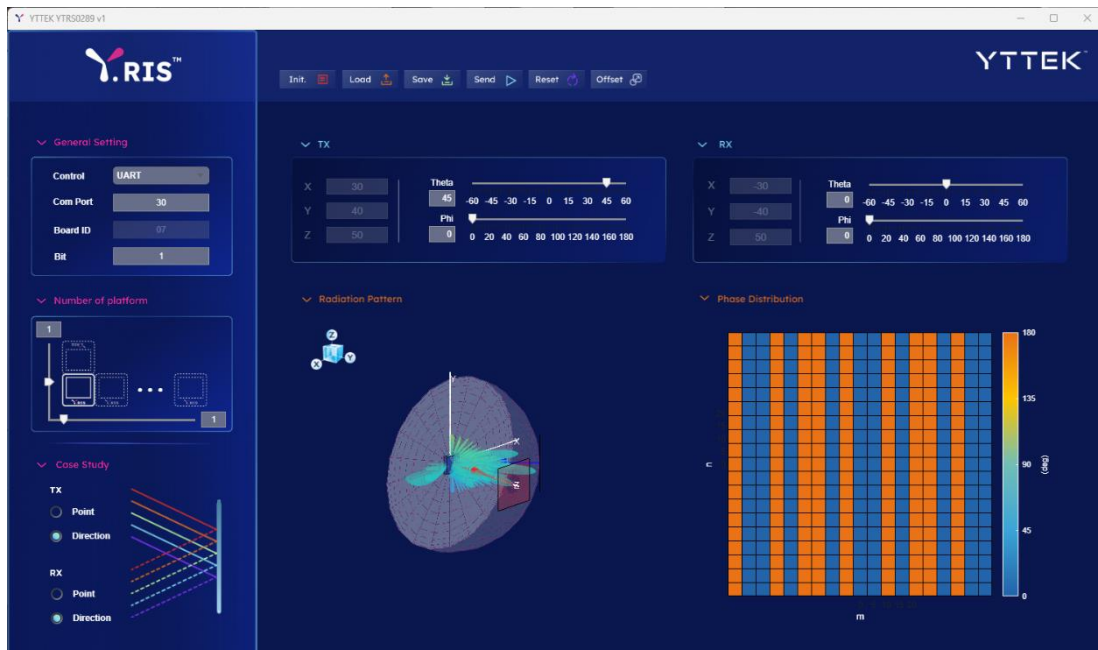
## 3.2 Graphical User Interface (GUI) Quick Start

In this section, we will provide a brief introduction to the use of the GUI. Users can perform system configuration directly through GUI.

- GUI path: “.\\release\\YTRS\_GUI\_v2 “



(a) Default



(b) Beam setting

Figure 3-5: YTRS0289 GUI

After opening the GUI, users can see the GUI window. Below, we will introduce how to operate it step by step, starting with the basic functions & control functions.



Figure 3-6: GUI function bar

- **INIT:** Initialize the Y-FORCE.
- **LOAD:** Read previously saved files.
- **SAVE:** Save the current settings.
- **SEND:** After setting the parameters, the user needs to click "SEND" to transmit the control signal to YTRS0289.
- **RESET:** Reset GUI mode.
- **Offset:** Add 90 degrees to the current phase setting. The user does not need to click "SEND" again; the change is applied immediately.

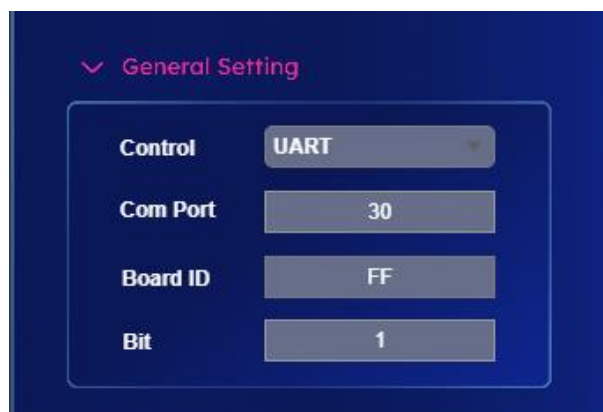


Figure 3-7: Control function bar

- **Control:** Choose how to control the YTRS0289, either via UART or DCI.
- **Com Port:** COM port value when using UART control. You can find the COM port value in the Device Manager.
- **Board ID:** YTRS0289's ID.
- **Bit:** set the resolution of phase [Unit: bit]

We provide four RIS transmission modes for users to choose from according to their needs.

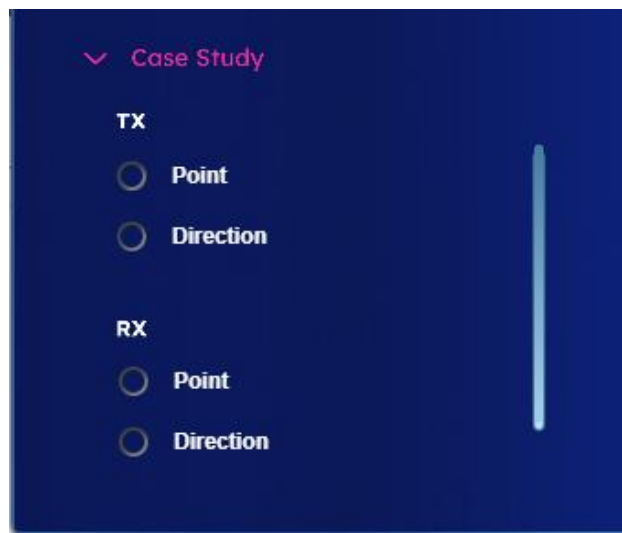
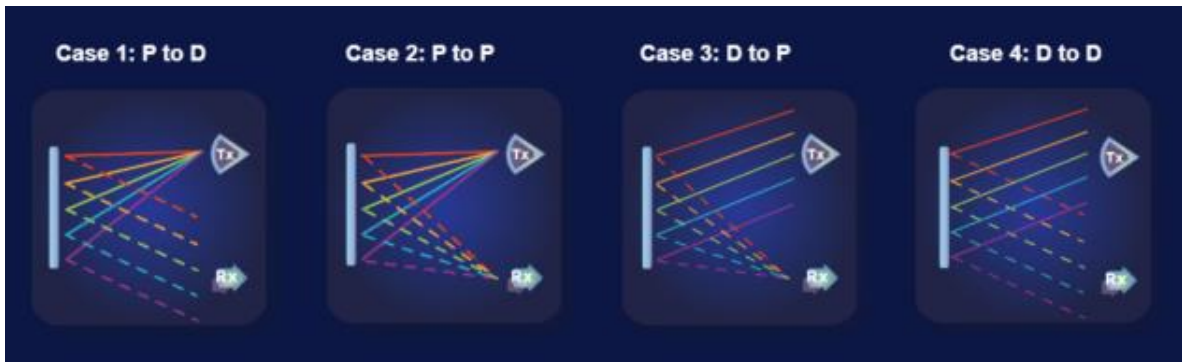


Figure 3-8 GUI mode setting

- **TX Point:** A “point source” is incident on the RIS.
- **TX Direction:** A “plane wave” is incident on the RIS.
- **RX Point:** Reflect the signal towards a specific “point”.
- **RX Direction:** Reflect the signal as a “plane wave”.

After selecting the transmission mode, users can configure the positions and angles of both TX and RX here. The positions or angles of transmitter and receiver are according to the coordinate shows in below figure (unit: mm).

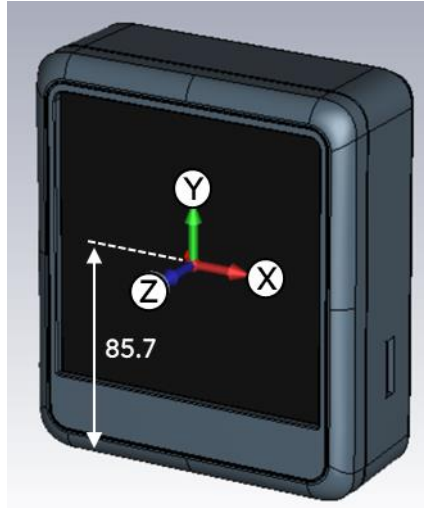


Figure 3-9: YTRS0289 coordinate

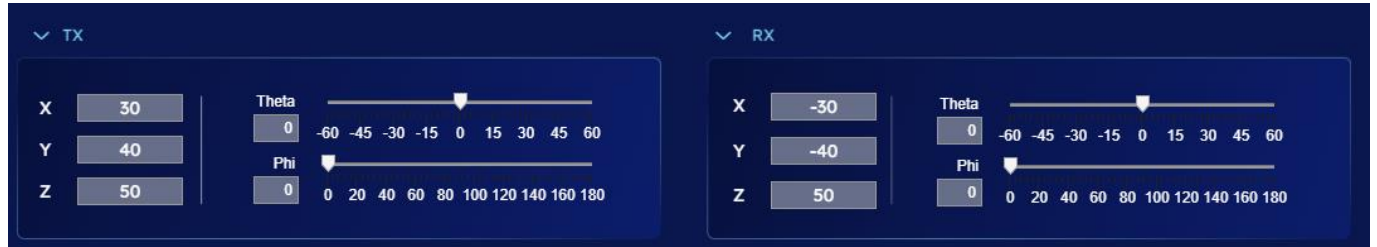


Figure 3-10: GUI TX&RX position setting

- **Origin** of coordinate is at the center of the panel.
- **X-axis** is the horizontal axis of the RIS surface. (unit: mm)
- **Y-axis** is the vertical axis of the RIS surface.
- **Z-axis** is the axis perpendicular to the RIS surface.
- **Theta** represents the polar angle between the Z-axis and radial line. (unit: deg)
- **Phi** represents the azimuthal angle between X-axis and the radial line's project onto the XY plane.

User can see a more intuitive depiction of the signal transmission status in the two images below. In this example, we are using “case2” with TX coordinates of  $X=30$ ,  $Y=40$ ,  $Z=50$ , and RX coordinates of  $X=-30$ ,  $Y=-40$ ,  $Z=50$ . In figure 3-11, users can easily view the radiation pattern illustrating the signal is transmitted through the RIS to the RX. In figure 3-12, user can see the phase distribution of each cell on the RIS.

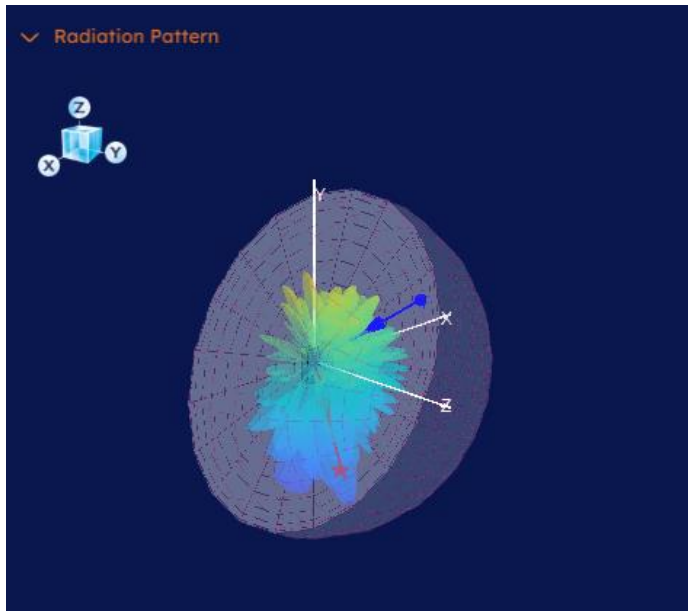


Figure 3-11: GUI RIS radiation pattern

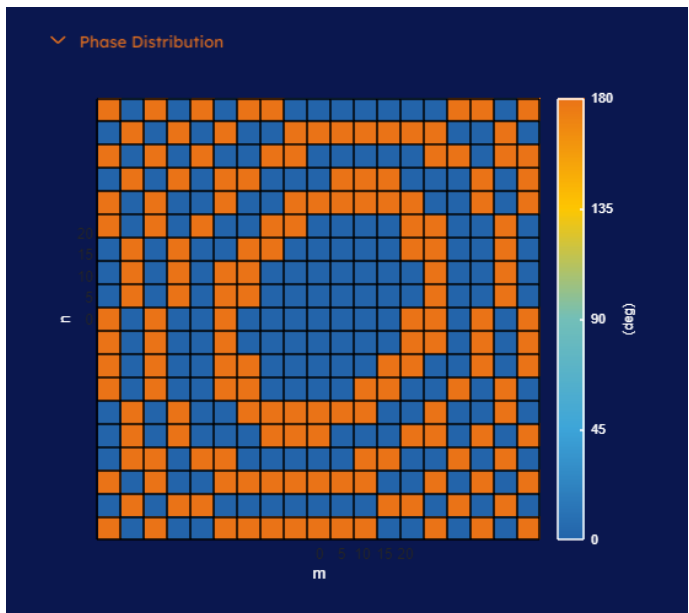


Figure 3-12: GUI RIS phase distribution

### 3.3 Introduction of MATLAB APIs

This section will introduce all the APIs function of YTRS0289. **All APIs are in the folder “.\YTRS0289\”**. User can easily control RIS to execute the experiments you want by using the APIs on YTRS0289. (The detail introduction of APIs in chapter 3.3.2 and 3.3.3)

#### 3.3.1 Initialize (DCI Control)

First, when the Y-FORCE power is turned on, enter “Init(type)” in the command window to initialize the Y-FORCE platform.

- Type → 0: all platform are single, 1: Master slave mode. (If type is not entered, the default is 0.)

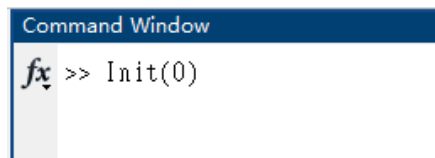


Figure 3-13: Initial Command

If the initialization is complete, the console would show the message “Software version:2.5.2456, HW Version: xx” and “Initial Completed! ”.

The folder “Tool” contains all APIs for users. The related path of “Tool” will be added after platform be initialized.

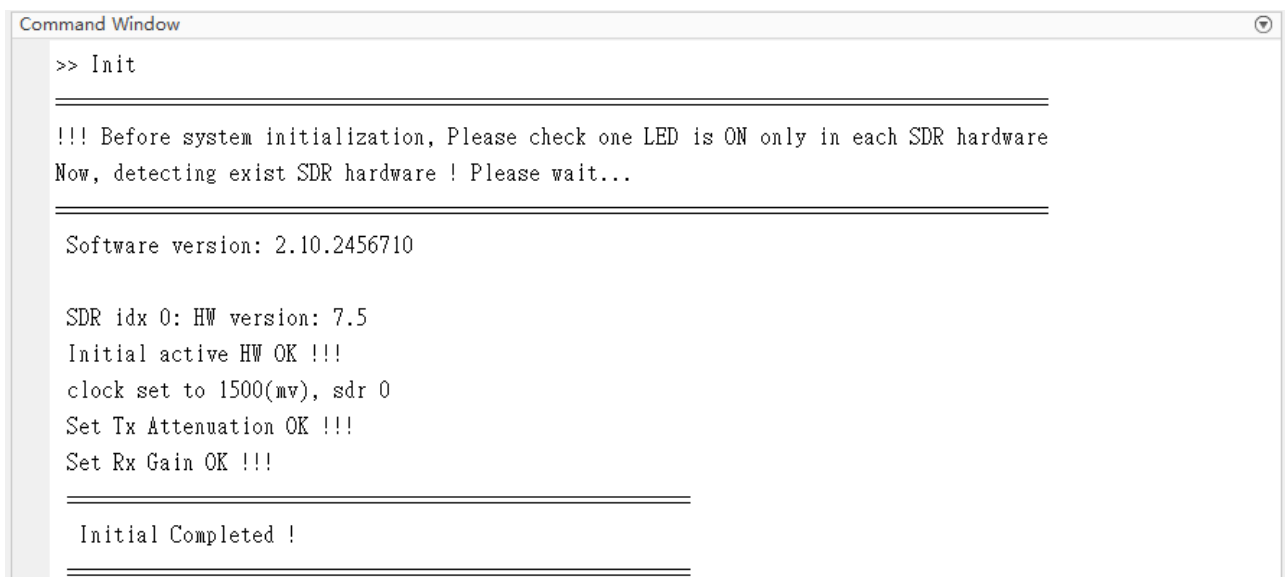


Figure 3-14 Initial Message

### 3.3.2 Parameters Setting

1. **RIS\_Case**: according to the position of transmitter and receiver choose an appropriate test case. (Figure 3-15 (a))
  - **Case 1**: Reflected point source towards plane wave.
  - **Case 2**: Reflected point source towards point.
  - **Case 3**: Reflected plane wave towards point.
  - **Case 4**: Reflected plane wave towards plane wave.
2. **Freq**: select the operation frequency between 29.2GHz-29.6GHz. (Figure 3-15 (b))
3. **(x,y,z) & (theta,phi)**: set the positions or angles of transmitter and receiver. (Figure 3-15 (b))
4. **Bits**: set the resolution of phase. (Figure 3-15 (c))
5. **(num\_px,num\_py)**: set the number of panels. (Figure 3-15 (c))
6. **Id\_panel**: input the ID of YTRS0289 is expressed in hexadecimal. (Figure 3-15 (c), please refer to the chapter 4.3)
7. **Fig**: configure the figure of phase distribution and array factor. (Figure 3-15 (c))
8. **Idx\_SDR**: choose the index of Y-FORCE. (Figure 3-15 (c)) (Refer to the chapter 3.3.1)

```

1      %% Main Test for YTRS
2
3      addpath('./YTRS0289');
4      % Test Case:
5      % Case 1: Point to Direction
6      % Case 2: Point to Point
7      % Case 3: Direction to Point
8      % Case 4: Direction to Direction
9
10     % Test Case
11     RIS_Case = 2;
12

```

(a)

```

13      %% Parameter Setting
14      % Freq
15      freq = 29.4; % unit: GHz
16
17      % TX
18      xi = 0;      % unit: mm
19      yi = 0;      % unit: mm
20      zi = 0;      % unit: mm
21
22      % RX
23      xr = 0;      % unit: mm
24      yr = 0;      % unit: mm
25      zr = 0;      % unit: mm
26
27      % TX
28      theta_i = 0;    % unit: degree
29      phi_i = 0;      % unit: degree
30
31      % RX
32      theta_r = 0;    % unit: degree
33      phi_r = 0;      % unit: degree

```

(b)

```

35      % Normalize bits
36      bits = 1;      % Range: 1~11
37
38      % Panel
39      num_px = 1;
40      num_py = 1; % MAX. = 2;
41
42      id_panel = '00'; % RIS ID (HEX)
43
44      % Figure
45      % fig = 0, Off
46      % fig = 1, Plot phase distribution
47      % fig = 2, Calculate and plot array factor
48      % fig = 3, Both
49      fig = 0;
50
51      % SDR Index
52      idx_SDR = 0;

```

(c)

Figure 3-15 API Parameters Setting

### 3.3.3 MATLAB API introduction

In this section, we explore four phase distribution calculation APIs in detail, including Point to Direction(P to D), Point to Point(P to P), Direction to Point(D to P) and Direction to Direction(D to D) four cases. Those cases determine the behavior of the incident signal in relation to reflected signal. In the case of “Direction(D)”, the signal must have a specific angle for direction, either incident or reflection. In the “Point(P)”, the signal need to precisely positioned for either incident or reflection.

```

55  %% Calculate Phase Distribution
56  switch RIS_Case
57      case 1
58          % P to D
59          [phs_out,phs_re]=cal_ris_phs_p2d(freq,xi,yi,zi,theta_r,phi_r,bits,num_px,num_py,fig);
60      case 2
61          % P to P
62          [phs_out,phs_re]=cal_ris_phs_p2p(freq,xi,yi,zi,xr,yr,zr,bits,num_px,num_py,fig);
63      case 3
64          % D to P
65          [phs_out,phs_re]=cal_ris_phs_d2p(freq,theta_i,phi_i,xr,yr,zr,bits,num_px,num_py,fig);
66      case 4
67          % D to D
68          [phs_out,phs_re]=cal_ris_phs_d2d(freq,theta_i,phi_i,theta_r,phi_r,bits,num_px,num_py,fig);
69      otherwise
70          % P to D
71          [phs_out,phs_re]=cal_ris_phs_p2d(freq,xi,yi,zi,theta_r,phi_r,bits,num_px,num_py,fig);
72  end
73
74
75  %%
76  num_panel = num_px*num_py;
77
78  for np = 1:num_panel
79      % Transform Phase into Voltage
80      % vol_out: 20x20
81      [vol_out] = phs2vol(phs_out(:, :, np), freq);
82
83      % Send Voltage distribution to YTRS
84      id_in = id_panel(np, :);
85      Send_vol_to_YTRS(idx_SDR, id_in, vol_out);
86  end

```

Figure 3-16 Main Function of APIs

## I. Case1 : P to D Phase Distribution

- **[Phs,Phs\_re] =**  
**cal\_ris\_phs\_p2d(freq,xi,yi,zi,theta\_r,phi\_r,bits,num\_px,num\_py,fig)**
- This function can calculate the phase on RIS, while the RIS reflects signal from a point source to RX direction.

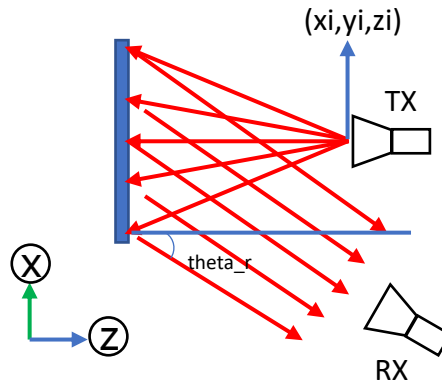


Figure 3-17: transmission scenario Point to Direction

**Phs:** The phase distribution on RIS. [Unit: deg.]

**Phs\_re:** The phase distribution of the reflected signal. [Unit: deg.]

**freq:** Center frequency. [Unit: GHz]

**xi,yi,zi:** The phase center of the transmitter. [Unit: mm]

**theta\_r:** The theta angle of the reflected signal. [Unit: deg., Range: -60~60]

**phi\_r:** The phi angle of the reflected signal. [Unit: deg., Range: 0~180]

**bits:** Phase step control [Unit: bit]

**num\_px:** The number of panel at the x direction

**num\_py:** The number of panel at the y direction

**fig:** Configure the figure of phase distribution and array factor

## II. Case2 : P to P Phase Distribution

- **[Phs,Phs\_re] = cal\_ris\_phs\_p2p(freq,xi,yi,zi,xr,yr,zr,bits,num\_px,num\_py,fig)**
- This function can calculate the phase on RIS, while the RIS reflects and focuses signal from a point source to RX point.

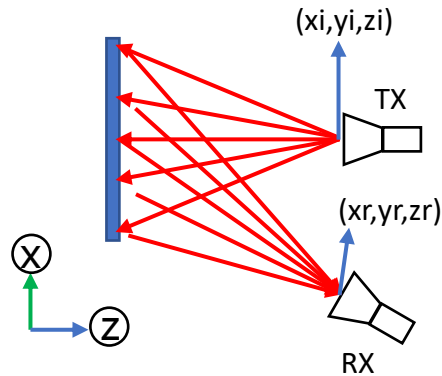


Figure 3-18: transmission scenario Point to Point

**Phs:** The phase distribution on RIS. [Unit: deg.]

**Phs\_re:** The phase distribution of the reflected signal. [Unit: deg.]

**freq:** Center frequency. [Unit: GHz]

**xi,yi,zi:** The phase center of the transmitter. [Unit: mm]

**xr,yr,zr:** The focal point the reflected signal [Unit: mm]

**bits:** Phase step control [Unit: bit]

**num\_px:** The number of panel at the x direction

**num\_py:** The number of panel at the y direction

**fig:** Configure the figure of phase distribution and array factor

### III. Case3 : D to P Phase Distribution

- **[Phs,Phs\_re] =**  
**cal\_ris\_phs\_d2p(freq,theta\_i,phi\_i,xr,yr,zr,bits,num\_px,num\_py,fig)**
- This function can calculate the phase on RIS, while the RIS reflects and focuses plane wave to RX point.

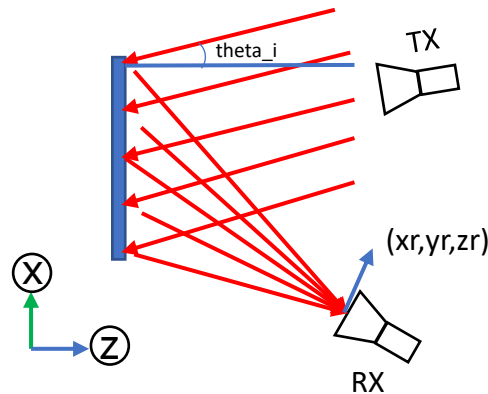


Figure 3-19: transmission scenario Direction to Point

**Phs:** The phase distribution on RIS. [Unit: deg.]

**Phs\_re:** The phase distribution of the reflected signal. [Unit: deg.]

**freq:** Center frequency. [Unit: GHz]

**theta\_i:** The theta angle of the incident signal. [Unit: deg., Range: -60~60]

**phi\_i:** The phi angle of the incident signal. [Unit: deg., Range: 0~180]

**xr,yr,zr:** The focal point the reflected signal [Unit: mm]

**bits:** Phase step control [Unit: bit]

**num\_px:** The number of panel at the x direction

**num\_py:** The number of panel at the y direction

**fig:** Configure the figure of phase distribution and array factor

#### IV. Case4 : D to D Phase Distribution

- **[Phs,Phs\_re] = cal\_ris\_phs\_d2d(freq,theta\_i,phi\_i,theta\_r,phi\_r,bits,num\_px,num\_py,fig)**
- This function can calculate the phase on RIS, while the RIS reflects plane wave to RX direction.

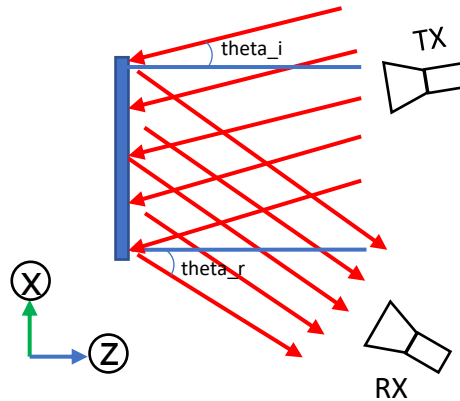


Figure 3-20: transmission scenario Direction to Direction

**Phs:** The phase distribution on RIS. [Unit: deg.]

**Phs\_re:** The phase distribution of the reflected signal. [Unit: deg.]

**freq:** Center frequency. [Unit: GHz]

**theta\_i:** The theta angle of the incident signal. [Unit: deg., Range: -60~60]

**phi\_i:** The phi angle of the incident signal. [Unit: deg., Range: 0~180]

**theta\_r:** The theta angle of the reflected signal. [Unit: deg., Range: -60~60]

**phi\_r:** The phi angle of the reflected signal. [Unit: deg., Range: 0~180]

**bits:** Phase step control [Unit: bit]

**num\_px:** The number of panel at the x direction

**num\_py:** The number of panel at the y direction

**fig:** Configure the figure of phase distribution and array factor

#### V. Convert Phase to Voltage

- **[vol\_out] = phs2vol(phs\_t1, freq)**
- This function can convert every cell's phase distribution to voltage distribution on RIS.

**vol\_out:** The voltage lean on the liquid crystal makes signal' phase shift on RIS. [Unit: Volt]

**phs\_t1:** The phase distribution on RIS. [Unit: deg.]

**freq:** Center frequency. [Unit: GHz]

#### VI. Write command to YTRS via **DCI**

- **Send\_vol\_to\_YTRS(indx\_SDR, id\_in, vol\_out)**
- This function can write voltage information to YTRS via DCI control.

**indx\_SDR:** YTPC-series platform ID [range: 0~3]

**id\_in:** The ID of YTRS0289 [range: 00~FF]

**vol\_out:** Every cell phase corresponds to voltage. [Unit: Volt]

#### VII. Write command to YTRS via **UART**

- **RIS\_CMD\_Write\_Frame(id\_in, vol\_out, my\_COM)**
- This function can write voltage information to YTRS via UART control.

**id\_in:** The ID of YTRS0289 [range: 00~FF]

**vol\_out:** Every cell phase corresponds to voltage. [Unit: Volt]

**my\_COM:** The COM port number used by the user.

## 4 Test Procedure

### 4.1 Test Environment

We use an OTA (over the air) system to test the performance of YTRS0289. The location of transmitter is at the **45°** and the receiver's is at the **0°**.

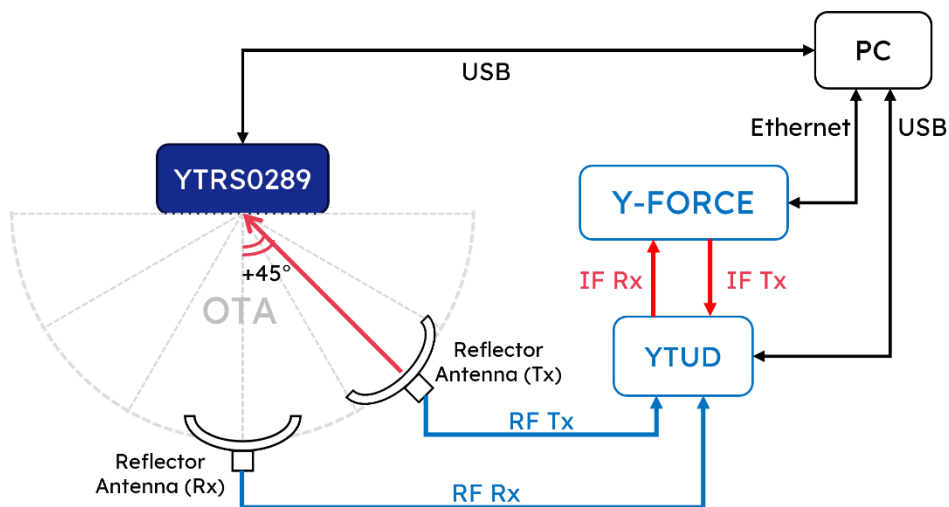


Figure 4-1: Block diagram of test environment



Figure 4-2: Test environment

## 4.2 GUI Setting Procedure (UART Control)

### Step1

1. GUI path: “.release\YTRS\_GUI\_v2 “
2. Execute the program **YTTEK\_YTRS\_V2.exe**
3. There are two windows, the GUI of YTRS and the command window.

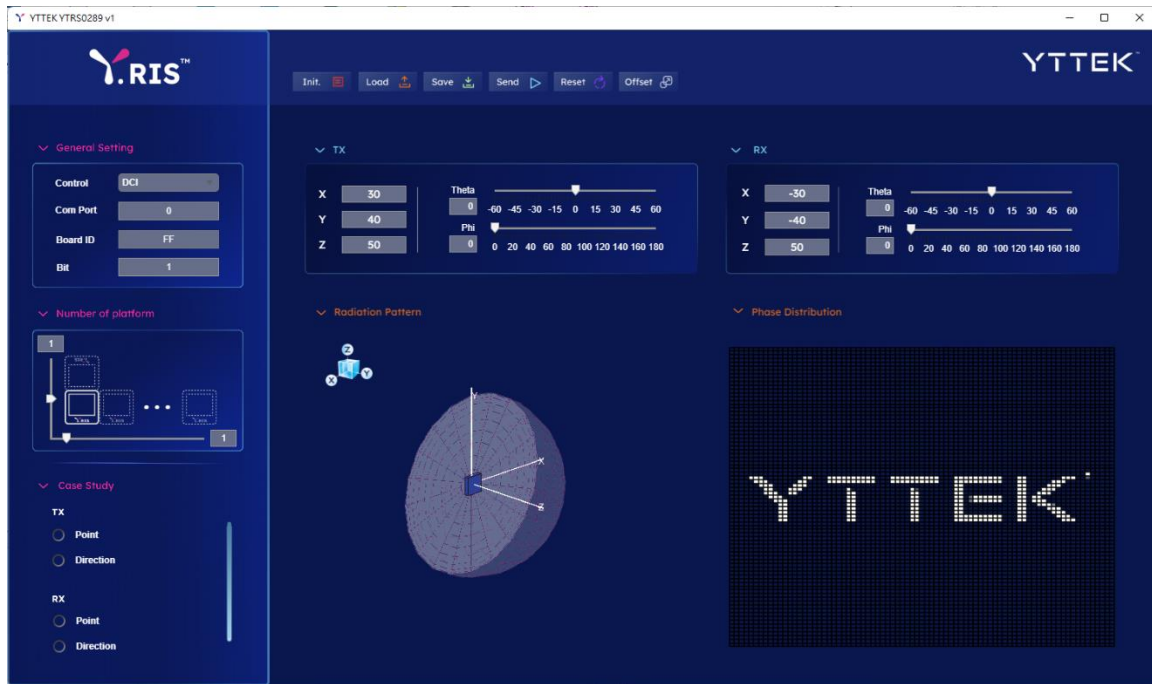


Figure 4-3: YTRS0289 GUI

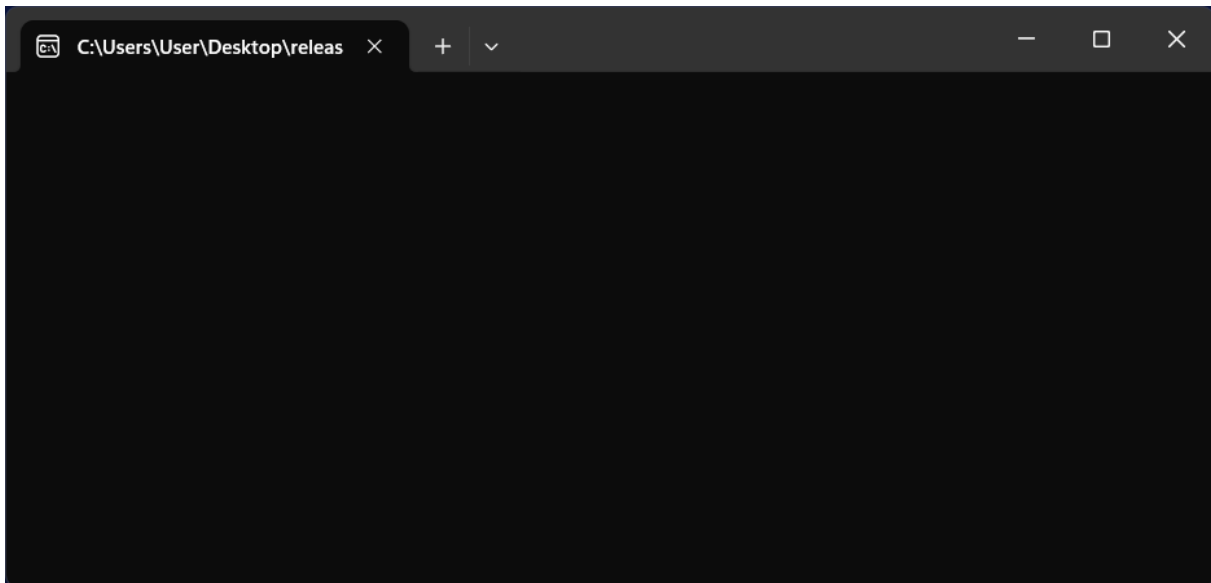


Figure 4-4: Command window

## Step2

1. Choose the **Control** mode as **UART**.
2. Enter the **COM Port** number(user can obtain the COM port number through Device Manager)
3. Click the **INIT** button to get **Board ID** and the command window will show the message.

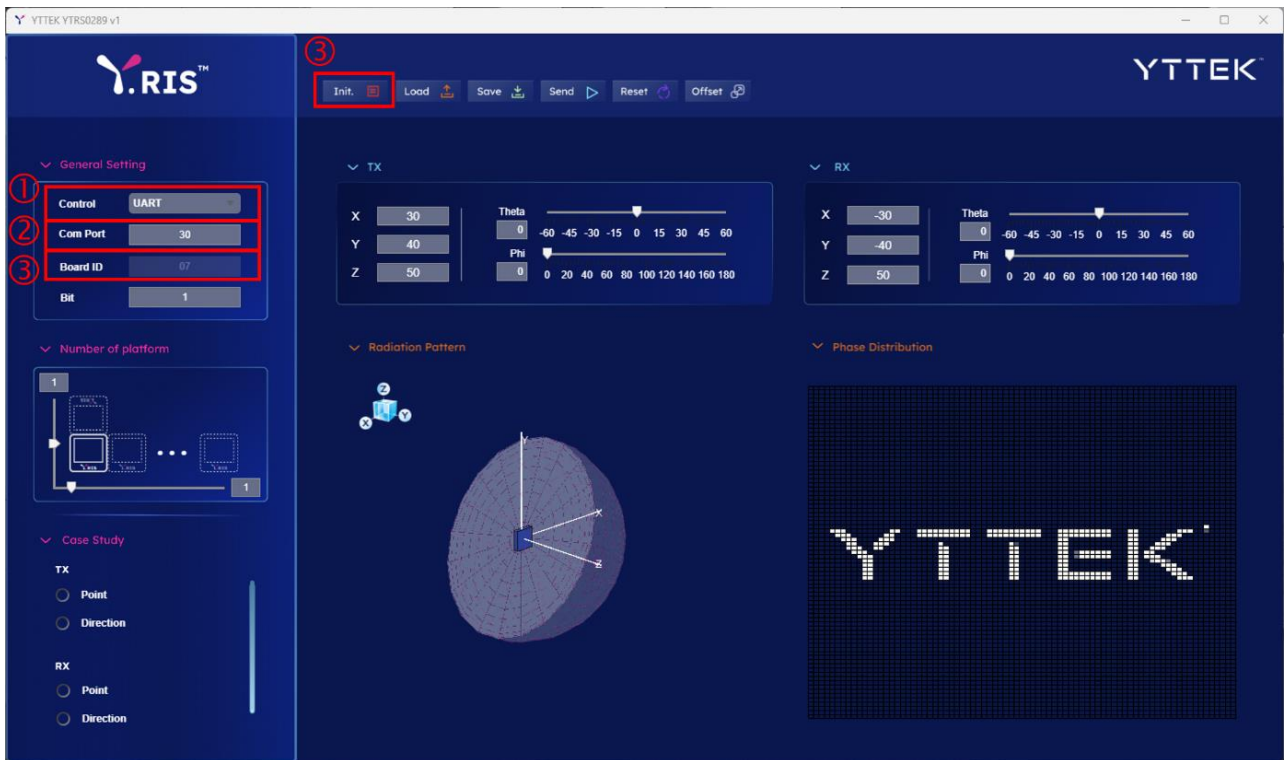


Figure 4-5: GUI with INIT button

```

read_value =

    0

loop_num =

    1

id_panel =

    '07'

Open Com Port Finish!!
loop_now=2
Success.
Close Com Port Finish!!
Done.

read_value =

    0
  
```

Figure 4-6: Get Board ID completed

### Step3

1. Choose the **Case:4 D to D**. (please refer to the chapter 4.3)
2. Input **45°** and **0°** into **Theta** and **Phi** of transmitter.
3. Input **0°** and **0°** into **Theta** and **Phi** of receiver.
4. Input **1** into **Bit**.

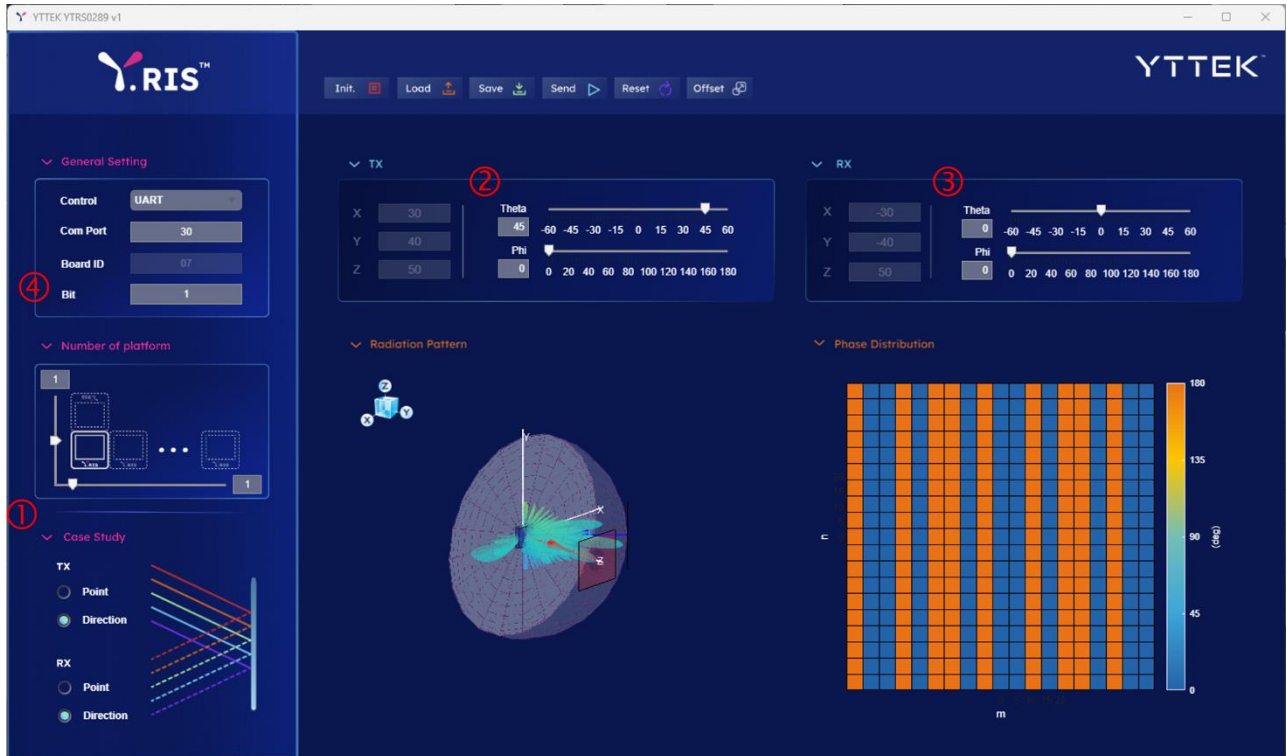


Figure 4-7: Parameters of transmitter and receiver

## Step4

1. Click the **SEND** button and wait the voltage distribution send to the YTRS0289.
2. The command window shows a **Success** message.



Figure 4-8: Parameters of transmitter and receiver

```

RIS_Case =

    4

CRC =

    'FEF8'

Open Com Port Finish!!
loop_now=6
Success.
Close Com Port Finish!!
Done.

read_value =

    0
  
```

Figure 4-9: Voltage setting completed

## Step5

1. Click the **Offset** button and the phase will add 90 degrees and send the voltage distribution to YTRS0289 immediately.
2. The command window shows a **Success** message.

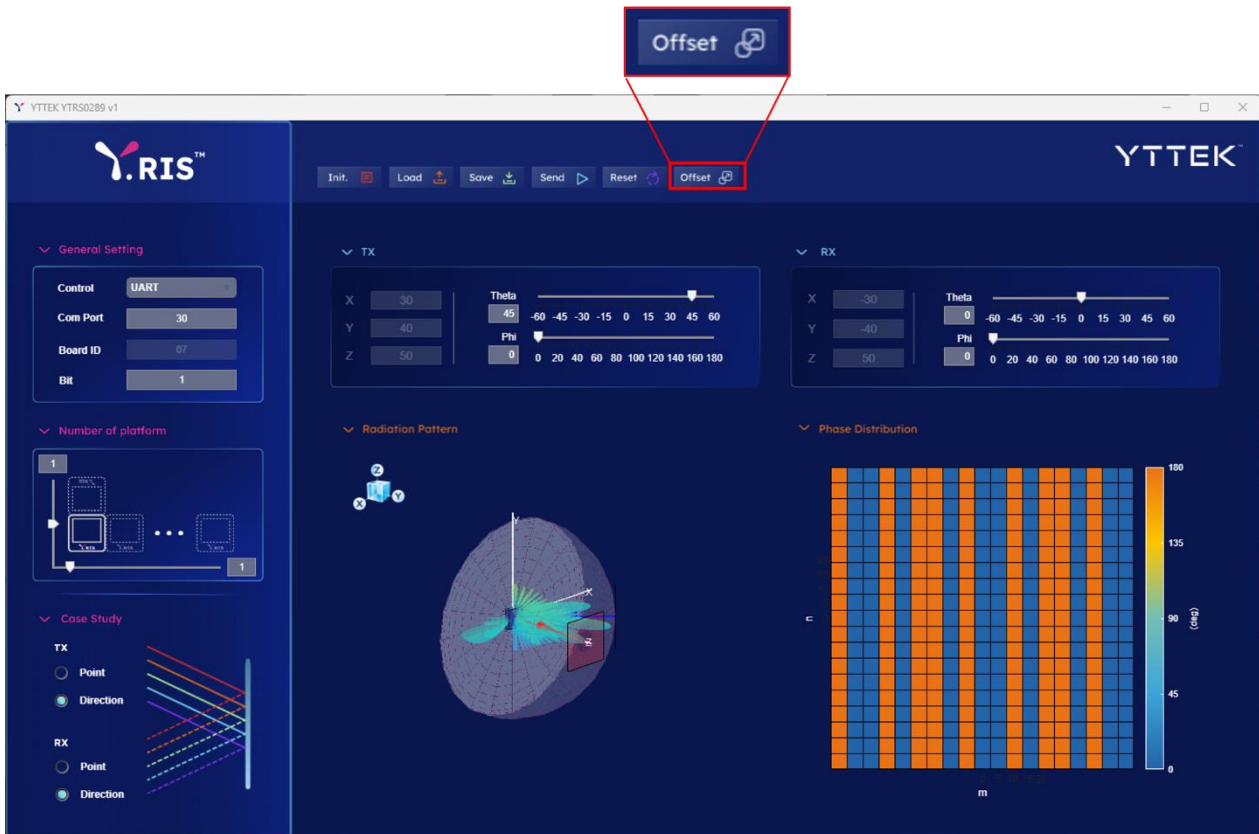


Figure 4-10: Parameters of transmitter and receiver

```
RIS_Case =
    4

CRC =
    'FFF0'

Open Com Port Finish!!
loop_now=6
Success.
Close Com Port Finish!!
Done.

read_value =
    0
```

Figure 4-11: Offset completed

## Step6

- (For operations related to Y-FORCE, please refer to the “**YTPC-series Platform Quick Start Guide**”)
- Finally, we use Y-FORCE to transmit and receive 16 QAM signal and the EVM (Error Vector Magnitude) decreases more than 7dB, while YTRS0289 is power on and steers beam to the receiver.

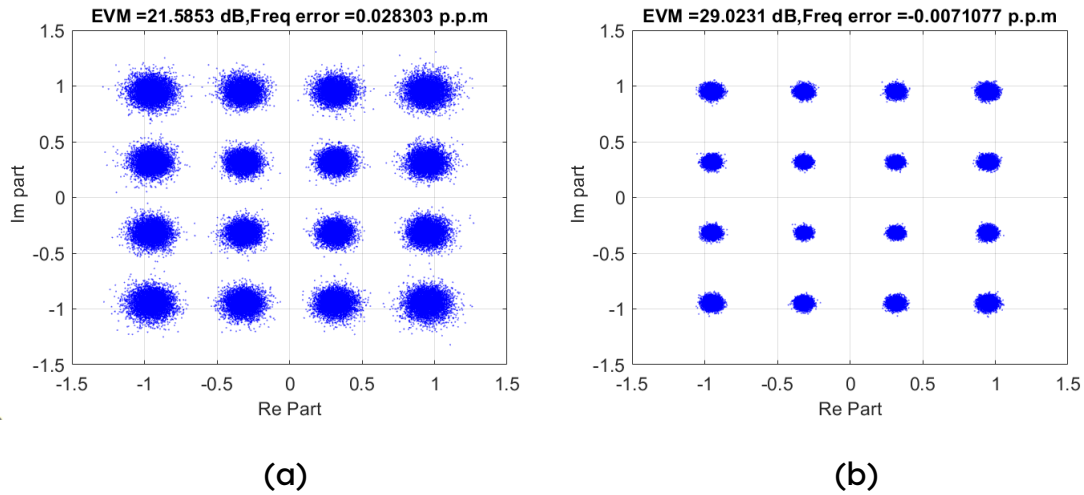


Figure 4-10: 16 QAM EVM (a) power off (b) steering beam

### 4.3 Appendix (Select YTRS0289's ID)

The YTRS0289's ID is used to distinguish each device when cascade plurality of YTRS0289. The YTRS0289 receives the command and checks the ID from the command is whether matching to the hardware. If the ID is not matching, the YTRS0289 will not execute the action of command. The hardware ID is controlled by an 8-bit DIP switch which provides the number "01~FE". (The IDs "00" and "FF" have been used for other functions.) Please refer to the below figure, an example of the DIP switch ID, and write the same ID into the MATLAB API or GUI before operating the YTRS0289.

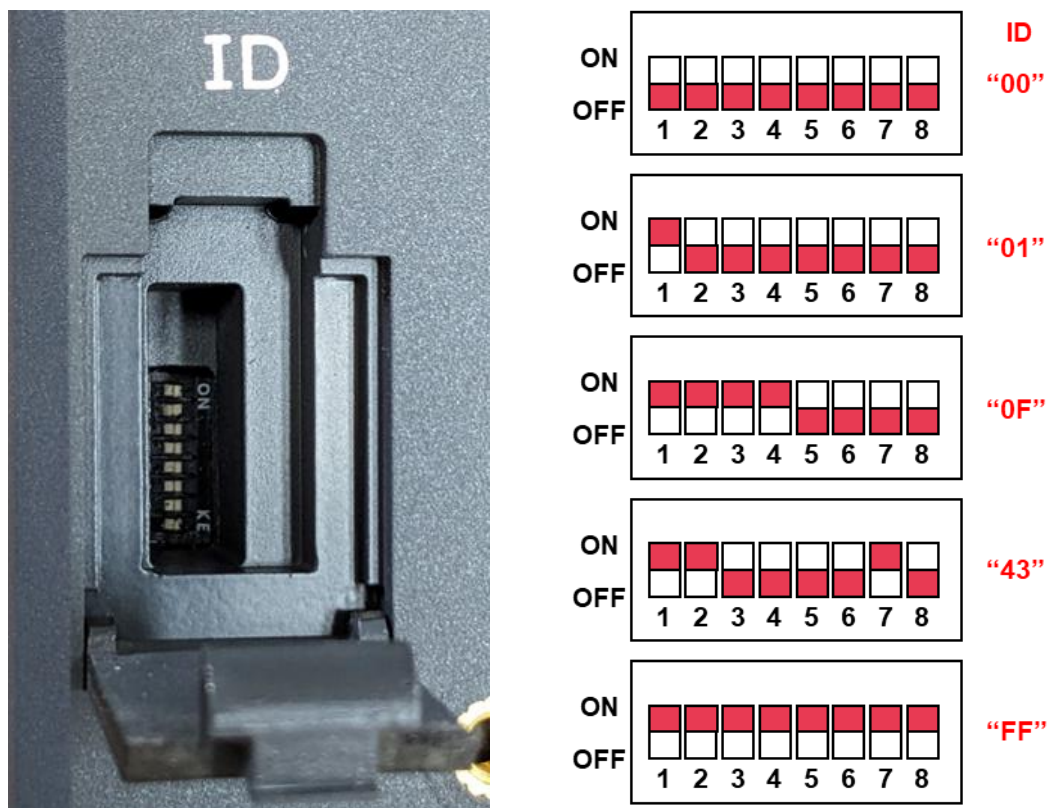


Figure 4-11: YTRS0289's ID Format

## **5 Troubleshooting**

## 6 Document Version History

| Version | Description                                                      | Author                        | Data       |
|---------|------------------------------------------------------------------|-------------------------------|------------|
| v1.0    | Deliver date: <b>2023/Aug</b><br>Modify file description details | HJ & Terry & Kevin<br>@ YTTEK | 2023/10/25 |
| v1.1    | Deliver date: <b>2024/Mar</b><br>Add Chapter for UART Control    | Terry & HJ @ YTTEK            | 2024/6/7   |
| v1.2    | Deliver date: <b>2024/Sept</b><br>Modify the picture of the GUI  | Terry @ YTTEK                 | 2024/9/30  |
|         |                                                                  |                               |            |